



Aeronautical Application Suite (AAS) Version Description Document

Foundation Sustainment Services (FS2)

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Revision History

Version #	Implemented By	Revision Date	Reason
1.2.170.3	Christi Curtis	11/25/2024	Initial release of this version of the product.

Authorization Notification

Any authorization applies only to the current version of the document.

Approval

The undersigned have read this VDD and approve its contents.

[INSERT NAME]

[Gov PM]

Date

[INSERT NAME]

[Contractor PM]

Date



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1 Scope

1.1 Introduction

This document is the *Version Description Document (VDD) for AAS Version 1.2.170.3*

This VDD details the software components to be delivered with AAS Version 1.2.170.3. This document describes the features included in the delivery of the software build, at entrance to test. It includes the build artifacts for deployment, any known issues with the delivery, and the changes included in the software version.

1.2 System Overview

The Aeronautical Application Suite (AAS) is a cloud-based upgrade that combined several stand-alone legacy Aeronautical desktop applications into a single, modernized cloud-based web production system. As part of the AAS, these upgraded applications are used to collect, update, store, maintain, and produce worldwide aeronautical data for a variety of military, civilian, and foreign customers.

The AAS Web Application is a web-based user interface for the AAS system. It interacts with the AAS Backend Application Programming Interface (API) to allow users to create, update, and delete aeronautical data. AAS Web application is written using Node.js and the React framework, and the source code files are compiled into a set of JavaScript, HTML, CSS, and image files that can be deployed on any web server software that can serve static content. AAS is accessed through a modern web browser on a client workstation.

The web interface utilizes React and serves as the user interface to the backend API developed as a cross-platform service using C# and .NET Core, which interacts with a modernized PostgreSQL database.

The AAS Backend Application is a C# application that provides a reliable and secure REST API for interacting with the AAS database. It enforces business logic and processing rules, ensuring that all changes to data are valid and consistent. The AAS Backend Application acts as the gatekeeper for all interactions with the database through Entity Framework ORM, and all client programs and external system connections must process their changes through the API.

1.3 Document Overview

The Aero AAS VDD provides the list of software delivered for Version 1.2.170.3 of the product and the capabilities of that software. This document details a single release of the product and will be updated only to correct information specific to version 1.2.170.3 of the product. Follow-on versions of the product will be documented in a separate VDD specific to that version of the product.

The document is organized as follows:

- Section 1 – Scope
- Section 2 – Version Description



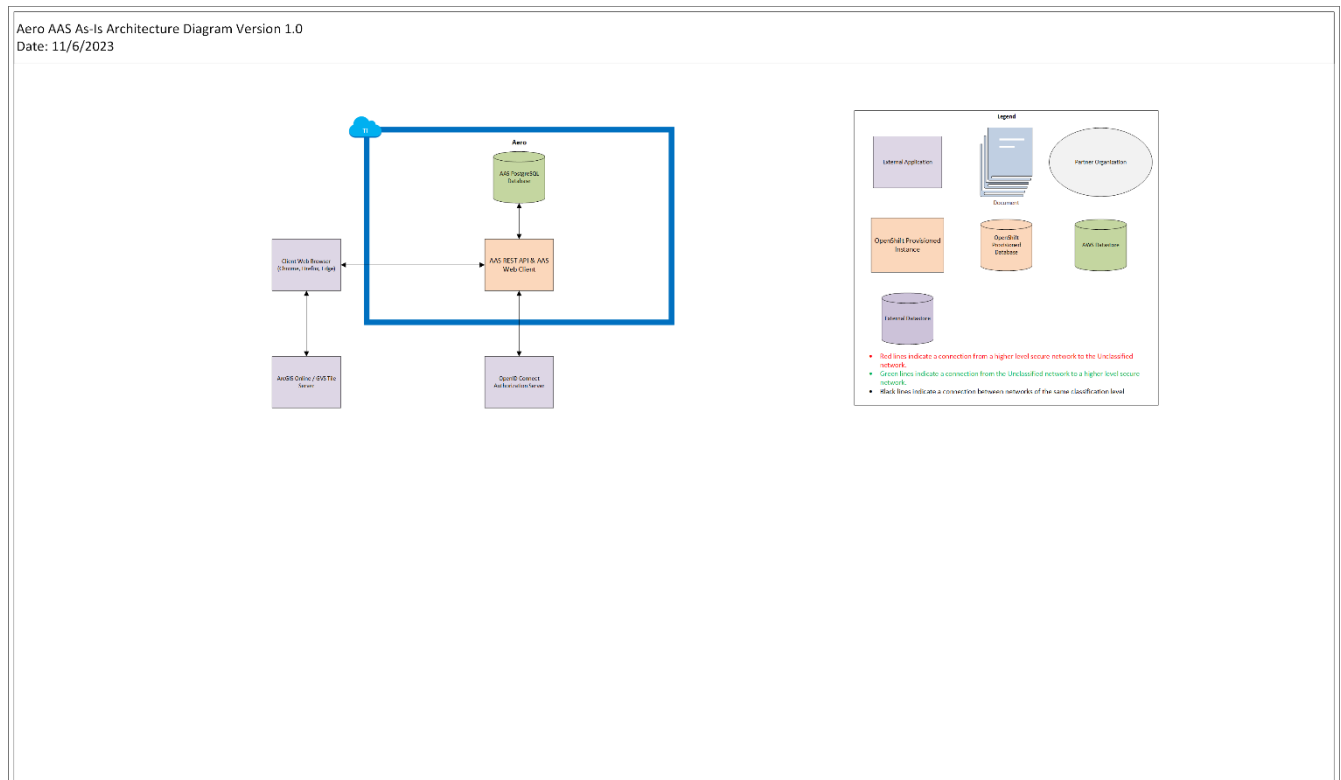
2 Version Description

This release represents an update to the existing AAS system, providing the same basic capabilities that exist in the current release. Several customer UAT corrections have been implemented along with code updates to ongoing development of features. Multiple NPM packages were upgraded after some became disapproved, and a few NuGet packages were updated to fix a vulnerability that was discovered in scans. This release is also the first in which spatial features have been merged into the database.

2.1 Deployment Environment

The AAS infrastructure will be deployed to the NGA Kubernetes Environment utilizing the OpenShift platform. Relational databases will be deployed using AWS Relational Database Service (RDS) instances. The AAS application components (server and web client) will be deployed to OpenShift pods using containers defined by the application team. The following diagram illustrates external connections to services and applications leveraged by the AAS infrastructure.

Architecture Diagram





2.2 Software Functionality and Capabilities

The AAS is a web-based user interface which interacts with the AAS Backend Application Programming Interface (API) to allow users to create, update, copy, and delete aeronautical data. An administration section allows administrators to manage users, groups, and permissions. An Outputs section allows users to import, create, verify and download AAFIF, DCDT, and DAFIF data.

2.3 Inventory of Software Components

This section documents the software configuration for AAS.

2.3.1 Software Deliverables

AAS will deliver custom software packages and build artifacts containing the application components, including but not limited to C# DLL files and JavaScript, HTML, and CSS scripts. These packages will include code library dependencies along with custom code to ensure that the application has all the software needed to execute.

2.3.2 Software List

The AAS Version 1.2.170.3 Software Lists for Custom code, COTS and Open Source code can be found in Table 2.3.2.

Table 2.3.2 - Software List

Software	Version	SWID	Vendor	Control #	Notes
Aeronautical Application Suite (AAS)	1.2.170.3	19829	Aeronautical Navigation Office (SFA)	2024120244	
UI NPM Packages					
buffer	6.0.3	24721	Feross Aboukhadijeh	20220222027	
core-js	3.37.0	7588	Denis Pushkarev	20240813007	
cross-fetch	4.0.0	21311	Leonardo Quixada	20230725063	
dayjs	1.11.10	21213	iamkun	20231019055	
domain-browser	4.22.0	28492	MIT	20230523063	
formik	2.4.2	20724	Jared Palmer	20230705009	
https-browserify	1.0.0	24726	Feross Aboukhadijeh	20220222022	
ka-table	8.3.2	28552	komarovalexander	20240625074	
leaflet	1.9.4	31327	Vladimir Agafonkin	20240402046	
leaflet-arrowheads	1.4.0	22628	Seth Lutske	20230523021	
leaflet-coordinates-control	1.0.2	28197	pedroalvesr	20230308086	
leaflet-draw	1.0.4	18575	Leaflet	20170124022	
leaflet-geometryutil	0.10.3	28200	Makina Corpus	20240625075	
react-leaflet-measure	3.1.0	27779	M. Hasbie	20230614034	
leaflet-polylinedecorator	1.6.0	25160	bbecquet	20220315012	
leaflet-textpath	1.2.3	28202	Makina Corpus	20230308096	
leaflet-touch-helper	2.0.0	27775	Per Liedman	20230308097	
leaflet-zoombox	0.5.2	15713	Conservation Biology Institute	20230523013	
lodash	4.17.21	7564	John-David Dalton	20210329011	



normalize.css	8.0.1	15544	Nicolas Gallagher	20190108016	
proj4	2.11.0	27236	Andreas Hocevar	20240625076	
proj4leaflet	1.0.2	18594	Kartena	20190207031	
prop-types	15.8.1	26615	Facebook	20220331016	
query-string	7.1.1	16601	Sindre Sorhus	20220523097	
react	17.0.2	7572	Dan Abramov	20210604009	
react-color	2.19.3	18808	Case Sandberg	20210310022	
react-data-table-component	7.5.4	23854	John Betancur	20240626018	
react-datepicker	4.18.0	18074	Martijn Russchen	20231025039	
react-dom	17.0.2	31097	Joseph Savona	20210719025	
react-hot-loader	4.13.1	18210	Dan Abramov	20240626022	
react-input-mask	2.0.4	24992	Nikita Lobachev	20220311017	
react-leaflet	4.2.1	14071	Paul Le Cam	20230328030	
react-leaflet-textpath	2.1.1	27795	Clement Allen	20230523011	
react-markdown-renderer	1.4.0	27786	Jonathon Mills-Kelly	20230309081	
react-modal-resizable-draggable	0.1.6	27784	Wesley Wang	20230523009	
react-redux	8.1.2	15720	Tim Dorr	20230829057	
react-router	5.3.4	13985	Tim Dorr	20230203038	
react-router-dom	5.3.4	23883	Tim Dorr	20230203040	
react-select	5.7.5	16020	Jed Watson	20240626026	
react-table-container	2.0.3	25466	Tanner Linsley	20220517040	
redux	4.2.1	14444	Dan Abramov	20230810026	
redux-thunk	2.4.2	23701	Redux	20230109032	
stream-http	3.2.0	24725	John Hiesey	20220222023	
styled-components	5.3.6	18548	Probablyup	20221004041	
node-url	0.11.3	24742	Roman Shtylman	20240606074	
yup	1.2.0	22145	Jason Quense	20230705025	
.Net (6) Packages					
Autofac	8.1.0	25477	Autofac	20241003078	
Autofac.Extensions.DependecnyInjection	10.0.0	25475	Autofac	20241003077	
AutoMapper	12.0.1	22280	Autofac	20230523060	
AutoMapper.Extensions.Microsoft.Depend encyInjection	12.0.0	27797	NuGet	20230523059	
AWSSDK.S3	3.7.307.17	20185	Amazon Web Services, Inc.	20240627040	
AWSSDK.SecurityToken	3.7.400.36	32353	NuGet	20241018072	
Castle.Core	5.1.1	16424	The Castle Project	20230725003	
CoordinateSharp	2.13.1.1	27799	NuGet	20230310026	
DocumentFormat.OpenXml	2.19.0	27800	NuGet	20230523055	
Microsoft.EntityFrameworkCore	6.0.35	25412	Microsoft	20241114120	
Microsoft.AspNetCore.Authentication.JwtB earer	6.0.35	25083	Microsoft	20241114118	
Microsoft.AspNetCore.SpaServices.Extens ions	6.0.35	27802	NuGet	20241114119	
NetTopologySuite.IO.GeoJSON4STJ	4.0.0	27803	NuGet	20241007117	
NetTopologySuite.IO.ShapeFile	2.1.0	27804	NuGet	20230310034	
Npgsql.EntityFrameworkCore.PostgreSQL	6.0.29	27805	NuGet	20241114121	
Npgsql.EntityFrameworkCore.PostgreSQL. NetTopologySuite	6.0.29	27806	NuGet	20241114123	
SharpKml	6.0.0	12793	Samuel Cragg	20230523045	



Swashbuckle.AspNetCore	6.5.0	25175	Richard Morris	20230213013	
Swashbuckle.AspNetCore.Swagger	6.5.0	27807	Nuget	20230523044	
System.Private.Uri	4.3.2	33471	Microsoft	20241114126	
System.Text.Encoding.CodePages	6.0.0	32768	Microsoft	20240910025	
System.Text.Encodings.Web	8.0.0	32715	MIT	20240904080	
System.Text.Json	8.0.5	32979	NuGet	20241114129	
NPM Development Dependencies					
@babel/core	7.24.4	18857	Nicolo Ribaud	20240425059	
@babel/eslint-parser	7.24.1	21659	The Babel Team	20240321010	
@babel/node	7.23.9	20085	The Babel Team	20240627037	
@babel/plugin-proposal-class-properties	7.18.6	18945	The Babel Team	20230501045	
@babel/plugin-proposal-object-rest-spread	7.8.3	18856	The Babel Team	20200304015	
@babel/plugin-transform-react-jsx	7.23.4	18183	Henry Zhu	20241003031	
@babel/plugin-transform-runtime	7.24.3	23859	The Babel Team	20240320045	
@babel/preset-env	7.24.8	18855	The Babel Team	20240718098	
@babel/preset-react	7.24.1	19014	The Babel Team	20240320092	
@types/react	18.3.1	18058	Nathan Shively-Sanders	20241003033	
@types/lodash	4.17.0	18407	Google, Inc.	20241003034	
acorn	8.11.3	20267	Marijn Haverbeke	20240627062	
ajv	8.12.0	15059	Evgeny Poberezkin	20231120035	
babel-jest	29.7.0	30230	Babel-jest	20231206038	
babel-loader	9.1.3	15661	Daniel Tschinder	20230808072	
babel-plugin-transform-react-remove-prop-types	0.4.24	19058	Olivier Tassinari	20230522015	
bufferutil	4.0.8	24274	Luigi Pinca	20240627061	
chai	4.3.10	7555	Jake Leur	20231102026	
cross-env	7.0.3	17442	Kent C. Dodds	20220127022	
css-loader	6.8.1	7589	Alexander Akait	20240102051	
cypress	13.3.0	18486	cypress.io	20231020012	
eslint	8.57.0	15393	Nicholas Zakas	20240320032	
eslint-config-airbnb	19.0.4	15561	AirBnB	20220215022	
eslint-import-resolver-webpack	0.13.7	15546	Jordan Harband	20240627059	
eslint-plugin-import	2.28.1	15560	Jordan Harband	20231003012	
eslint-plugin-jsx-a11y	6.7.1	15587	Ethan Cohen	20230214005	
eslint-plugin-react	7.33.2	15721	Jordan Harband	20231019059	
eslint-plugin-react-hooks	4.6.0	19480	Facebook	20220920008	
eslint-plugin-security	1.7.1	19592	Node Security	20230427014	
file-loader	6.2.0	7545	Alexander Akait	20201117031	
jest	29.7.0	15573	Simen Bekkus	20231019061	
jest-html-reporter	3.10.2	20390	Johan Hargne	20240627058	
jest-junit	16.0.0	26772	Jason Palmer	20231017068	
mini-css-extract-plugin	2.9.0	17006	Alexander Akait	20240627057	
mkdirp	3.0.1	13862	James Halliday	20230501070	
npm-run-all	4.1.5	17058	Toru Nagashima	20190711009	
react-test-renderer	18.3.1	15507	Facebook	20240627056	
rimraf	5.0.5	7617	Isaac Z. Schuelter	20240627054	
sass	1.75.0	20614	Natalie Weizenbaum	20240627050	
sass-loader	14.2.1	13738	Alexander Akait	20240627048	



style-loader	3.3.3	7578	Alexander Akait	20240627044	
url-loader	4.1.1	13743	Alexander Akait	20210922026	
webpack	5.95.0	7585	Tobias Koppers	20241001067	
webpack-cli	5.1.4	16949	Even Stensberg	20230705020	
webpack-dev-middleware	7.2.1	15498	Tobias Koppers	20240627047	
webpack-hot-middleware	2.26.1	15504	Glen Mailer	20240627046	
webpack-node-externals	3.0.0	22413	Liad Yosef	20220523054	
.NET 6 Development Dependencies					
coverlet.collector	6.0.0	29679	Toni Solarin-Sodara	20230919023	
JUnitXml.TestLogger	4.1.0	33470	NuGet	20241114130	
Microsoft.EntityFrameworkCore.InMemory	6.0.35	29671	Microsoft	20241114131	
Microsoft.NET.Test.Sdk	17.11.1	20204	Microsoft	20241114132	
Moq	4.20.72	32811	NuGet	20241114133	
NUnit	4.2.2	20340	Charlie Poole	20241114134	
NUnit.Analyzers	4.3.0	33469	mikkelbu	20241114137	
NUnit3TestAdapter	4.6.0	20341	Charlie Poole	20241114138	
Verify.NUnit	28.1.3	33468	simoncropp	20241114139	

2.4 Changes Installed

As was noted in Section 1.3, this is considered to be in initial release of the 1.2.170.3 product so changes from the previous version are not recorded in this VDD.

Table 2.4 includes defects recorded in testing as well as any other internally discovered defects that were resolved.

- Issue Types:
- Test Problem Report (TPR): DRs written by the NTO during testing
- System Problem Report (SPR): DRs written by problems found by the PMO, customer, or external systems.

Table 2.4 - Changes Installed

PBI/INC #	Key	Issue Type	Summary

2.5 Installation Instructions

The application components will be installed using the NGA CORE services GitLab CI/CD pipeline. No special install instructions will be needed once the pipeline is set up and configured.

(Instructions on how to manually install and configure the application, if needed for this document, are To Be Determined).



2.6 Possible Problems and Known Errors

Table 2.6 Version 1.2.170.3 Possible Problems and Known Errors

Issue ID	Summary	Description